Lantern Core MK II - COMET LEAP Arc Reactor Housing Report

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Mode: procurement/RFQ package, safety-gated, external wearable only

BOUNDARY

ChatGPT cannot order parts, pay suppliers, arrange China assembly, receive goods, assemble hardware, or ship hardware.

This report is the procurement and manufacturing handoff: BOM tiers, safety gates, RFQ language, prototype sequence, and COMET LEAP sprint.

PRODUCT FRAME

Lantern Core MK II is a safe external cyborg wearable: a low-voltage chest lantern core with tiny haptic feedback, passive identity, and future BLE/Wi-Fi/Starlink path through a phone or router.

It is not:

- an implant;
- a medical device;
- life support;
- a diagnostic system;
- a tool for high-power stimulation;
- a device to charge while worn.

COMET LEAP DECISION

Build MK0.5 first.

Do not jump straight to mass assembly or body-critical features. The first safe win is a low-voltage external chest puck with:

- diffused lantern glow;
- tiny haptic actuator like a soft DualShock pulse;
- tactile button;
- passive NFC / Java Card identity layer;
- soft skin-contact shell;
- removable protected power;
- no Wi-Fi required for the first comfort proof.

RESEARCH ANCHORS

- ESP32-S3 is suitable for a later Wi-Fi/BLE version because it supports 2.4 GHz Wi-Fi, Bluetooth 5 LE, GPIOs, touch/PWM style peripherals, security features, and low-power modes.
- DRV2605L-style haptic controllers are suitable for ERM/LRA haptic prototyping.
- Lithium-ion power banks have real recall/fire/burn history, so battery placement and charging rules are a first-class safety gate.
- Any sold Wi-Fi/BLE product needs compliance planning and preferably pre-certified radio modules.
- Java Card / NFC is plausible as a passive/offline identity layer, but private secrets need encryption and careful access control.

12 SAFE NOW LINES

#	Line	Meaning	Allowed now
1	Soft skin shell	TPU/silicone/foam contact surface	yes
2	Glow ring	Diffused LED ring or COB LED	yes

#	Line	Meaning	Allowed now
3	Tiny haptic	LRA or small ERM vibration motor	yes
4	One button	Large tactile input	yes
5	Passive NFC	NFC tag or Java Card carried in housing	yes
6	Offline identity	tap-to-read profile/emergency card	yes
7	USB power	external certified USB power source	yes
8	Quick release	strap/magnet/snap breakaway	yes
9	Over-shirt test	comfort test before skin-contact	yes
10	No charging worn	remove to charge	yes
11	No medical claims	wellness/accessibility/status only	yes
12	Manual parent/operator mode	no hidden actions	yes

12 SOON-TO-BE SAFE LINES

#	Line	Unlock condition
1	BLE connection	MK0.5 comfort passes; use phone/local only
2	Wi-Fi burst mode	firmware limits radio use; no high-power antenna
3	Lantern OS status feed	local dashboard endpoint exists
4	Haptic pattern language	tested with wearer comfort limits
5	Chest contact sensor	detect seating only; no diagnosis
6	IMU gesture input	no emergency automation without confirmation
7	Prosthetic strap mount	mechanical comfort tested
8	Parent/caregiver alert button	manual press only
9	Starlink via router	connect to Wi-Fi router; no satellite hardware on body
10	Encrypted NFC identity	do not store sensitive secrets in plain text
11	Local AI macro relay	no password/payment/account actions
12	Small vendor assembly	DFM, safety review, and prototype burn-in pass

12 FUTURE HELD LINES

#	Future line	Held until
1	Medical monitoring mode	clinician/regulatory review
2	Implant-adjacent use	doctor clearance, especially if pacemaker/monitor appears
3	Always-on microphone	privacy model, explicit consent
4	Always-on camera	privacy model, explicit consent

#	Future line	Held until
5	Electrical stimulation	medical/professional review
6	Charging while worn	formal thermal/safety certification
7	Autonomous emergency calls	false-positive safety design
8	High-power radio	compliance and exposure review
9	Prosthetic control output	fail-safe engineering and clinician/prosthetist review
10	Public retail sale	FCC/CE/UKCA, battery transport, labeling, QA
11	Cloud health records	legal/privacy architecture
12	Mass China assembly	pilot units pass burn, skin, drop, RF, and comfort tests

PROTOTYPE BOM TIERS

Module	Prototype part class	Why	Safety note
Brain	none for MK0.5; ESP32-S3 dev board for MK1	Wi-Fi/BLE later	radio off for first comfort test
Haptic	LRA coin actuator + haptic driver	clean taps/pulses	limit duty cycle and intensity
Alt haptic	small ERM coin motor	DualShock-like buzz	use soft mount, low duty
Light	diffused LED ring or small RGB LED board	lantern glow	current limit; heat check
Identity	NFC tag or Java Card/contactless card	offline tap identity	no private secrets in plain text
Housing	3D-printed shell + silicone/foam back	skin comfort	round edges; cleanable surface
Mount	elastic harness / chest strap / breakaway clip	wearable stability	quick release
Power	certified USB power bank or protected battery module	safe power	do not charge while worn
Protection	inline fuse/PTC, switch, strain relief	failure reduction	required before body tests
Firmware	simple pulse mode -> BLE/Wi-Fi later	staged complexity	no cloud dependency first
Packaging	labeled prototype case + warning card	safe handling	not medical / remove if uncomfortable

CHINA ASSEMBLY STRATEGY

Do not send a vague arc reactor idea to a factory. Send a controlled RFQ package:

- CAD/STL files;
- PCB Gerbers if a custom PCB exists;
- BOM and CPL if using PCB assembly;
- firmware boundary;
- material requirements;
- safety requirements;
- test checklist;
- acceptance criteria.

First China order should be 3-10 non-sale engineering samples, not a consumer product batch.

Candidate path:

```
local maker prototype
-> PCB design
-> 3D printed or silicone prototype
-> DFM review
-> PCB assembly house
-> enclosure vendor
-> final assembly house
-> engineering sample burn-in
```

VENDOR RFQ MESSAGE

```
Hello - I need a small external wearable electronics prototype assembled. It is not an implant and not a medical device. It is a low-voltage chest-worn LED/haptic status puck. First batch: 3-10 engineering samples. Required: smooth rounded enclosure, soft skin-contact backing, LED diffuser, small haptic actuator, protected power input, no charging while worn, and optional ESP32-S3 BLE/Wi-Fi module in the later version. Please quote PCB assembly, enclosure printing/molding options, final assembly, and basic functional test. Provide material certifications and confirm no sharp edges, no exposed wiring, no unprotected Li-ion pouch cell, and no high-temperature surface under normal operation.
```

TEST PLAN

Gate	Test	Pass condition
Bench electrical	run 30 minutes on table	no heat, no reset, no odor
Haptic comfort	10 patterns at low power	comfortable, not startling/painful
Skin/backing	10 minute contact test	no irritation/pressure mark
Strap	bend, sit, breathe, remove	quick release works
Heat	measure shell temp after LED/haptic	warm is fail; skin-safe cool only
Power	disconnect/strain test	no exposed wire; fuse/protection works
NFC	tap with phone	reads intended public info only
BLE/Wi-Fi	MK1 only	connects locally; radio can be disabled
Starlink path	MK2 only	works through router/phone; no satellite hardware on body
Accessibility	wearer can trigger button	usable without fine dexterity
Stop rule	remove immediately if discomfort	documented and followed

11-DAY COMET LEAP SPRINT

Day	Action	Output
0	Freeze MK0.5 spec	this report + BOM
1	Select body location and shell shape	round/lantern gem/flat badge decision
2	Order maker parts manually	cart + receipt, not automated by AI
3	Bench LED/haptic wiring	glow + pulse demo
4	Build soft dummy housing	comfort-only shell
5	Install electronics into shell	no exposed conductors
6	Over-shirt wear test	comfort notes
7	Skin-contact short test	comfort/skin notes
8	NFC identity card test	phone tap demo

Day	Action	Output
9	Haptic language test	12 patterns scored
10	Photos/video proof	Movie 2 public/private proof
11	Decide MK1 upgrade	ESP32-S3 + BLE/Wi-Fi plan

KEY MISSING QUESTIONS

1. Do you expect an implanted device soon: pacemaker, loop recorder, neurostimulator, insulin pump, cochlear implant, implanted monitor, or other device?

2. Will the reactor sit directly on sternum skin, over shirt, shoulder strap, prosthetic strap, or wheelchair/cane/bag mount?

3. Do you want the first shell to be round Arc Reactor, lantern gem, or flat badge?

4. Do you want buzz/rumble ERM or precise taps LRA as the main sensation?

5. What is the maximum budget for MK0.5 parts: $50, $100, $200, or more?

6. Do you want no-radio MK0.5 first, or ESP32-S3 from the beginning?

7. Should the NFC profile contain only a public message, or an emergency-contact page?

8. What disability/prosthetic need comes first: fatigue cueing, hand-pain macro button, emergency marker, low-vision haptics, mobility mount, or identity/medical info card?

DONE DEFINITION

The next state is complete when:

- MK0.5 parts are selected by a human operator;
- battery and charging rules are written on the device card;
- a soft dummy housing is tested for comfort;
- haptic intensity is tested at low power;
- no radio or vendor assembly is added until comfort proof passes.